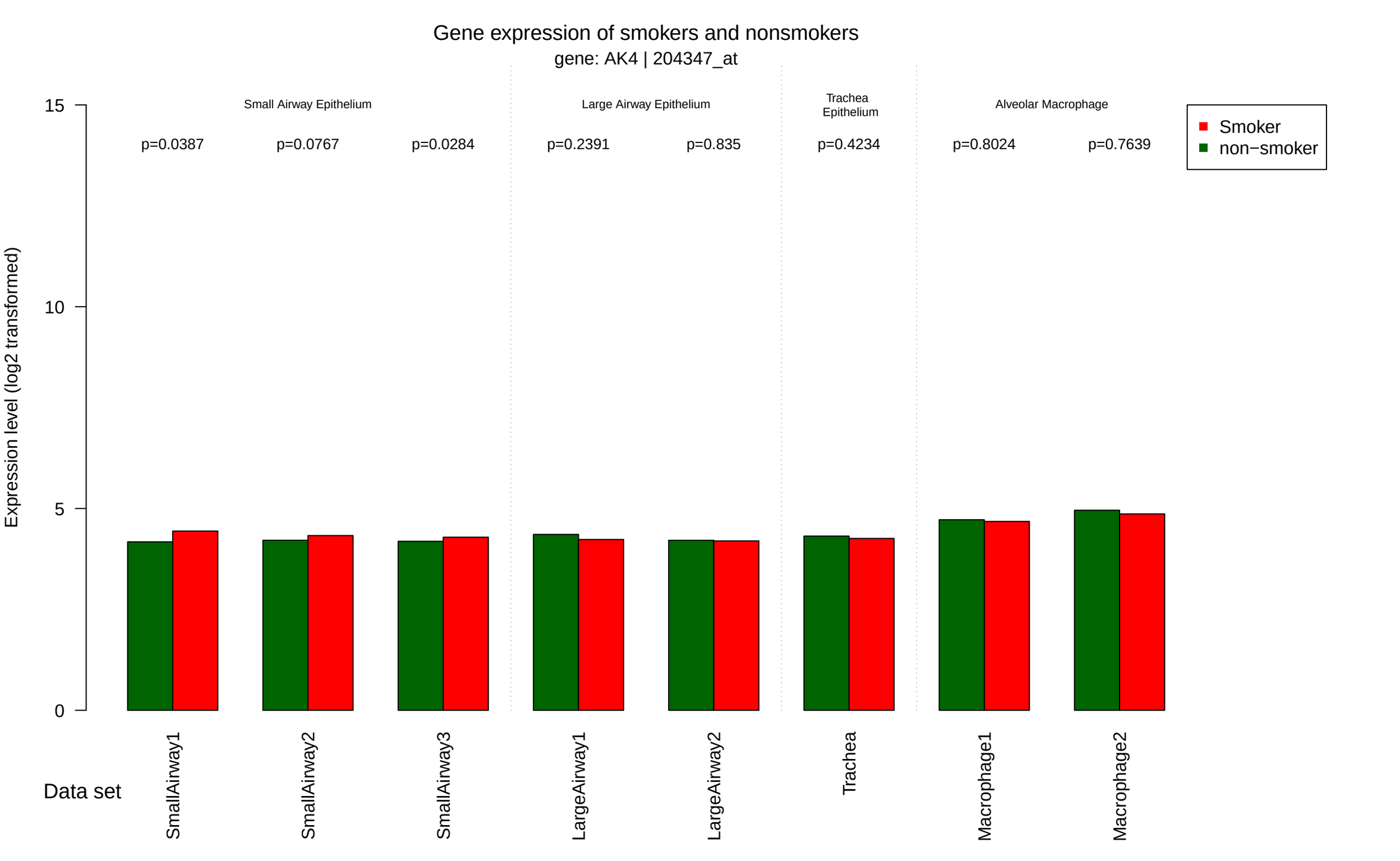
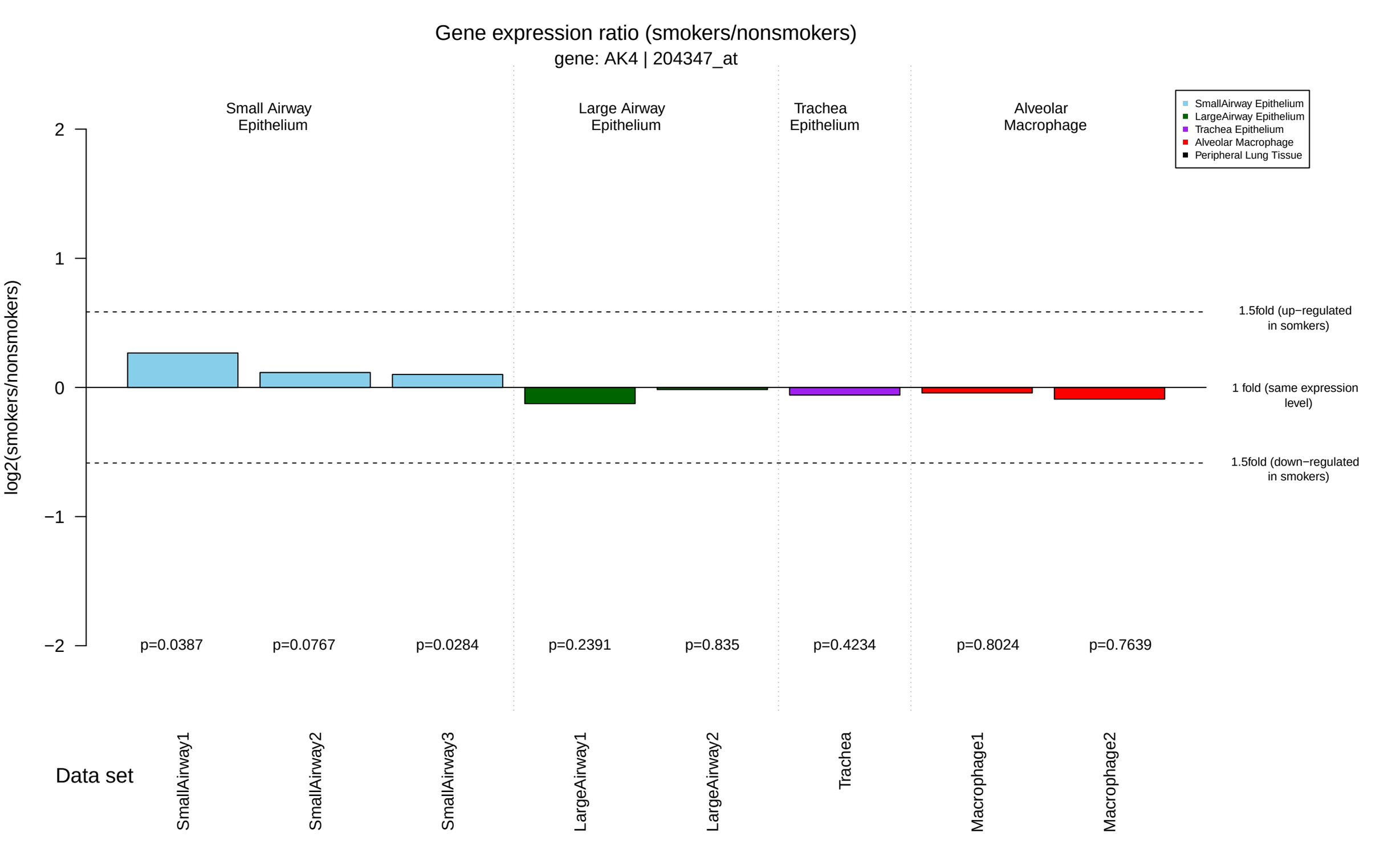
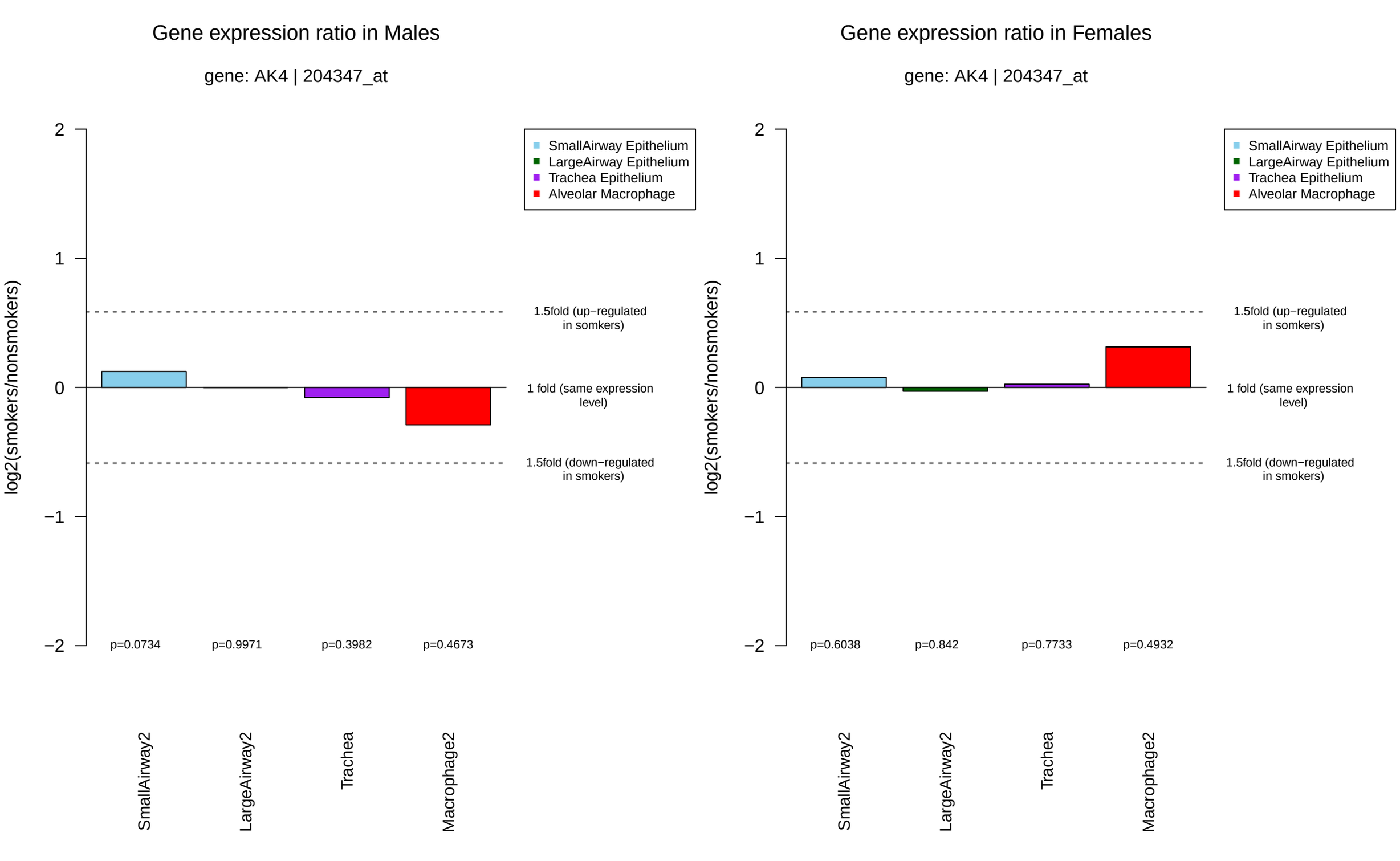
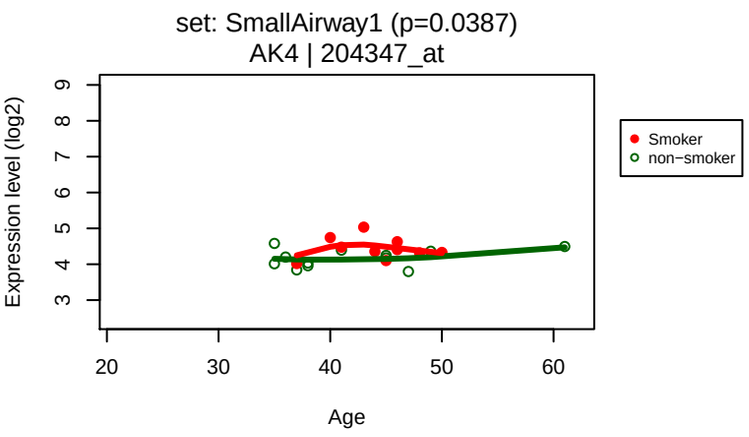
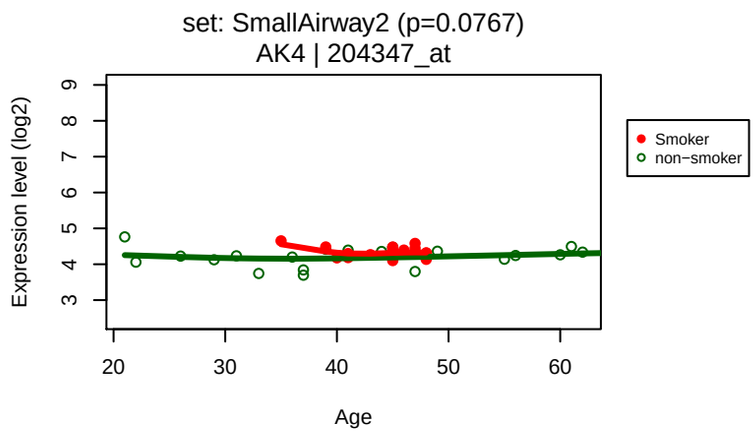
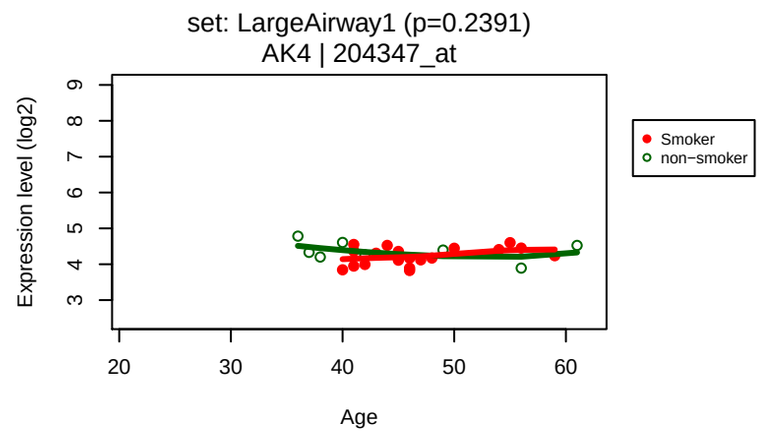

Gene expression of smokers and nonsmokers
gene: AK4 | 204347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0387
p=0.0767
p=0.0284
p=0.2391
p=0.835
p=0.4234
p=0.8024
p=0.7639
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

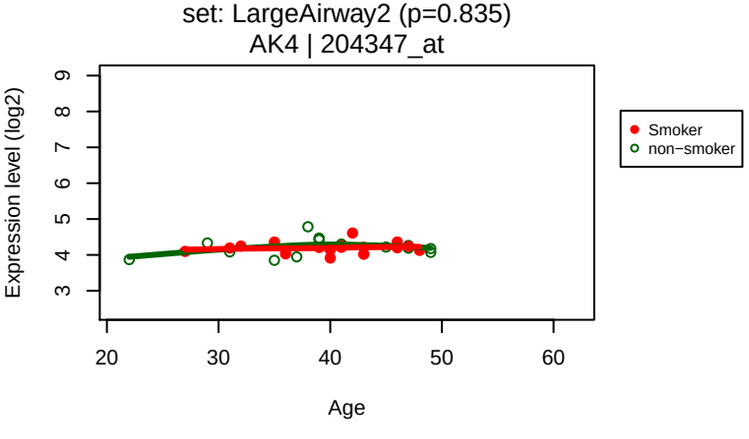

Gene expression ratio (smokers/nonsmokers)
gene: AK4 | 204347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0387
p=0.0767
p=0.0284
p=0.2391
p=0.835
p=0.4234
p=0.8024
p=0.7639
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

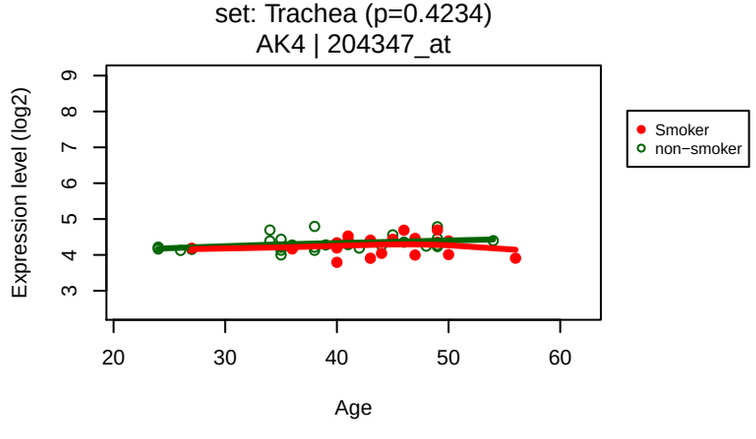

Gene expression ratio in Males
gene: AK4 | 204347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0734
p=0.9971
p=0.3982
p=0.4673
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK4 | 204347_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6038
p=0.842
p=0.7733
p=0.4932
SmallAirway2
LargeAirway2
Trachea
Macrophage2

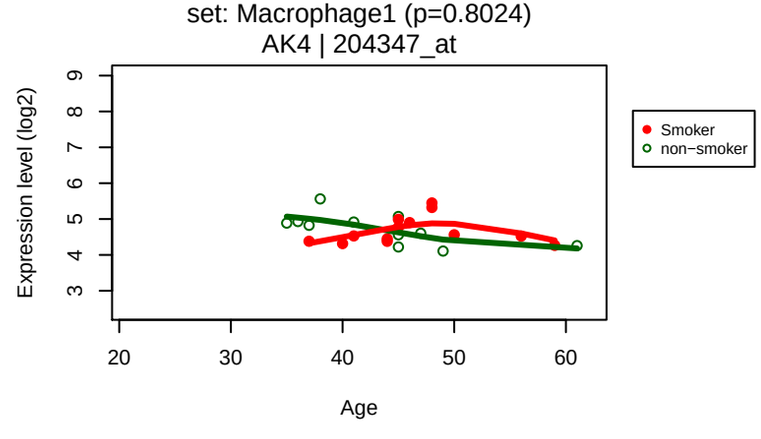

set: SmallAirway1 (p=0.0387)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker

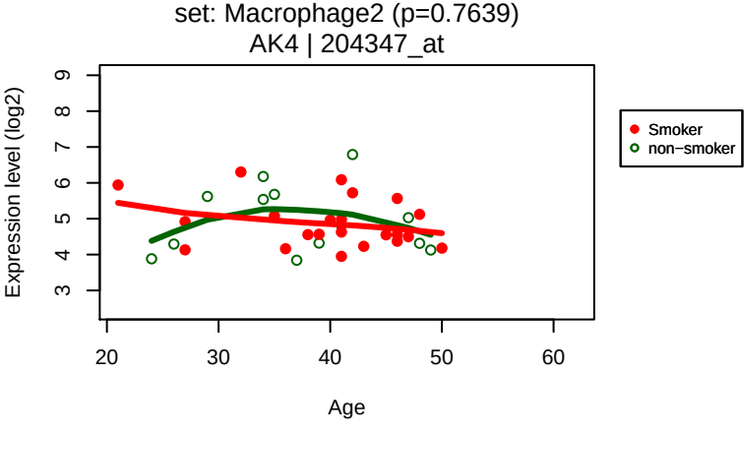

set: SmallAirway2 (p=0.0767)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker

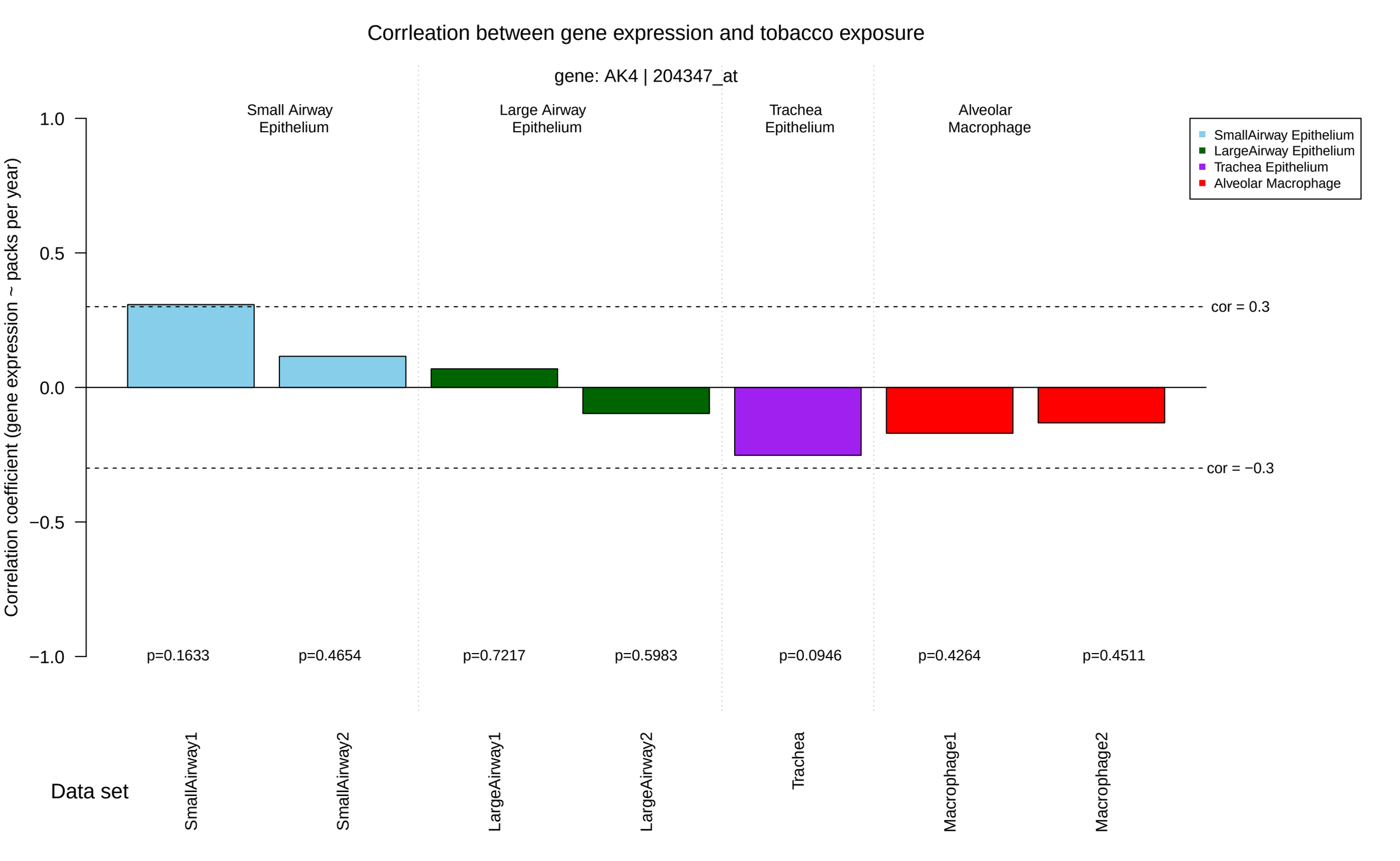

set: LargeAirway1 (p=0.2391)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.835)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4234)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8024)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7639)
AK4 | 204347_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK4 | 204347_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1633
p=0.4654
p=0.7217
p=0.5983
p=0.0946
p=0.4264
p=0.4511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2